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From Private Cognition to Collective Estimation: A Confidence-Aware Networked Consultation Framework for Heterogeneous Agents with Partial Information and Partial Reasoning

Zhishan Wang[1] and Rong Su[1]

***Abstract*— This paper studies collective estimation in multi-agent systems where each agent possesses only partial information and partial reasoning about a complex global process. We propose a self-aware social consultation framework over communication graphs. Each agent first constructs a private reasoning map from its local information and forms a private estimate of the global quantity. Its self-awareness is then updated according to historical estimation performance. During social consultation, agents exchange only belief-level information, and confidence-weighted aggregation is used to update their current estimates without modifying their local information sets or private reasoning maps. The framework is evaluated using real U.S. inflation data with heterogeneous agents and multiple communication structures. The results show that social consultation substantially improves both individual and population-level estimation accuracy, with consistent benefits across different network topologies and a historical out-of-sample period. Moreover, agents with weaker private cognition tend to obtain larger consultation gains. These results demonstrate that heterogeneous partial cognition can be exploited as a collective resource through self-aware belief-level communication.**

***Index Terms*— Multi-agent systems, distributed estimation, partial information, social learning, network of agents.**

I. INTRODUCTION

Complex systems are often difficult to describe accurately using a single mathematical model. In networked or multi-agent systems, local agents typically possess only partial information, while their reasoning is based on limited observations and past experience [1], [2]. For instance, in an economic system, no individual can obtain all information required to determine global economic changes; instead, each agent forms its own estimate from locally available information. Similar situations arise in sociological and large-scale networked systems, where individual beliefs and behaviors evolve through local interactions [3], [4]. This motivates the question of whether partially informed and partially reasoning agents can improve their estimation of a common global quantity through communication and information exchange.

Distributed estimation and consensus have been widely studied in networked systems [5], [6]. Related work has also considered state and fault estimation over complex networks [7], [8]. Information aggregation and selection under networked or limited-information settings have received considerable attention as well [9], [10]. Other studies have examined information-flow relationships and networked optimization [11], [12], while recent data-driven approaches have addressed heterogeneous multi-agent systems with unknown dynamics [13]. In parallel, opinion dynamics and social learning have investigated how beliefs and influence evolve through network interactions [14]–[16]. These studies commonly rely on explicit system, observation, or belief models, or focus primarily on information aggregation and convergence. Here, we consider heterogeneous agents with incomplete information and imperfect local reasoning, and study how historically realized reliability can guide belief-level communication when the global relationship itself is difficult to characterize.

In this paper, we propose a framework in which each agent learns a private reasoning map from its local information and forms a private estimate. Based on historical prediction errors, the agent updates its self-awareness to reflect the reliability and confidence of its private cognition. During social consultation, agents exchange beliefs and reliability information over a communication graph, and each receiver updates its estimate using confidence-weighted neighbor information. Importantly, consultation changes only the current belief, while the local information set and private reasoning map of each agent remain unchanged.

The main contributions of this paper are threefold:

1) We formulate a multi-agent estimation setting in which each agent possesses both partial information and partial reasoning, without requiring explicit knowledge of the unknown global relationship.
2) We introduce a self-aware belief-level consultation mechanism in which historical private performance determines social influence, and we characterize the local error condition under which social correction improves an agent's instantaneous estimate.
3) We validate the framework using a real-data U.S. inflation estimation testbed with 30 heterogeneous agents and 22 economic signals. Social consultation substantially improves both individual and population-level estimation, while all four tested communication structures outperform the private-cognition baseline.

The remainder of this paper presents the problem formulation in Section II, the self-aware social consultation mechanism in Section III, the experimental evaluation in Section IV, and conclusions in Section V.

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II. PROBLEM FORMULATION

A. Complex System and Partial Information

Let $\mathcal{S} := \{1, \dots, M\}$ denote the index set of globally observable system signals. Together, these signals form the global observable information vector. Let $\mathbf{x}_t = [x_{1,t}, \dots, x_{M,t}]^\top \in \mathbb{R}^M$ collect all globally observable signals. Consider a discrete-time system indexed by $t \in \mathcal{T}$. Let $y_t \in \mathbb{R}$ denote the global quantity of interest. We represent the unknown system relationship between the observable signals and the global quantity as $F_t^\star(\mathbf{x})$

$$y_t = F_t^\star(\mathbf{x}_t) + \varepsilon_t, \tag{1}$$

$F_t^\star(\mathbf{x})$ at this point does not represent a system that we can fully understand or explain; ε_t represents the uncertainty within this relationship. It merely serves as an analytical representation that may potentially correlate the observable signals with the global quantity. We may interpret $F_t^\star(\mathbf{x}) = \mathbb{E}[y_t \mid \mathbf{x}_t = \mathbf{x}]$, and we can say $\mathbb{E}[\varepsilon_t \mid \mathbf{x}_t] = 0$. This paper is not about how to identify or recognize $F_t^\star(\mathbf{x})$, but rather about how to estimate the global quantity y_t using the partial information available to the individual agent. At this point, we do not need to analyze the full system mechanism or relationships.

Let $\mathcal{V} := \{1, \dots, N\}$ denote the agent set. For each agent $i \in \mathcal{V}$, let $\mathcal{I}_i \subsetneq \mathcal{S}$ denote the subset of signals directly available to the agent, and let $m_i := |\mathcal{I}_i| < M$.

$$\mathbf{x}_{i,t} = \mathbf{P}_i \mathbf{x}_t \in \mathbb{R}^{m_i}, \tag{2}$$

Where $\mathbf{P}_i \in \{0,1\}^{m_i \times M}$ is the selection matrix, which indicates the types of information that each agent can obtain. In our model, different agents can observe overlapping signals. However, the population is heterogeneous, meaning at least some agents have different local information. The population is heterogeneous in that $\exists\, i \neq j$ such that $\mathcal{I}_i \neq \mathcal{I}_j$.

We assume the agent's local information is informative but individually incomplete. We assume $\mathrm{Var}(\mathbb{E}[y_t \mid \mathbf{x}_{i,t}]) > 0$ for all $i \in \mathcal{V}$. This condition indicates that once the agent acquires local information, it will modify its conditional estimate for y_t. Therefore, different agents, after obtaining different combinations of information, may capture different informative aspects or insights of the same global process. Information alone, however, does not specify how an agent interprets its observations.

B. Partial Reasoning and Private Cognition

We define $G_{i,t}^\star(\mathbf{z})$ as the ideal local conditional relation for agent i, and $G_{i,t}^\star(\mathbf{z})$ represents the best conditional relationship learning and inference for the global quantity that agent i can achieve based on its own local information.

$$G_{i,t}^\star(\mathbf{z}) := \mathbb{E}\left[y_t \mid \mathbf{x}_{i,t} = \mathbf{z}\right], \qquad \mathbf{z} \in \mathbb{R}^{m_i}. \tag{3}$$

Under squared loss, $G_{i,t}^*$ is the minimum-MSE predictor available to agent i when only $\mathbf{x}_{i,t}$ is observed. Even under perfect statistical knowledge, the agent can in general recover only $G_{i,t}^*$, rather than F_t^*. In reality, achieving $G_{i,t}^*$ is an ideal situation that the agent cannot know in practice. In practice, the agent can only learn and correct based on historical data $\mathcal{D}_i^{\mathrm{tr}}$.

Let $\mathcal{T}_{\mathrm{tr}} \subset \mathcal{T}$ denote the training period. Agent i learns from $\mathcal{D}_i^{\mathrm{tr}} := \{(\mathbf{x}_{i,\tau}, y_\tau) : \tau \in \mathcal{T}_{\mathrm{tr}}\}$ a reasoning map or a reasoning rule $f_i : \mathbb{R}^{m_i} \to \mathbb{R}$. We use the term *partial reasoning* to emphasize that f_i is learned only from the agent's restricted and incomplete information and finite historical experience or data, and is not assumed to be identical to the ideal local relation $G_{i,t}^*$.

We define the private cognition of agent i as $\mathcal{C}_i^P := (\mathcal{I}_i, f_i)$, which jointly describes what the agent observes and how it reasons from those observations. And we define the private estimate as:

$$\hat{y}_{i,t}^P = f_i(\mathbf{x}_{i,t}). \tag{4}$$

This is the estimate from the reasoning rule generated by the agent's historical data and experience.

We define the private error as:

$$e_{i,t}^P := y_t - \hat{y}_{i,t}^P. \tag{5}$$

In order to better describe and analyze the private estimate and error, we define the information-induced discrepancy $\delta_{i,t}^I$ and the reasoning-induced discrepancy $\delta_{i,t}^R$ as:

$$\begin{aligned} \delta_{i,t}^I &:= F_t^\star(\mathbf{x}_t) - G_{i,t}^\star(\mathbf{x}_{i,t}), \\ \delta_{i,t}^R &:= G_{i,t}^\star(\mathbf{x}_{i,t}) - f_i(\mathbf{x}_{i,t}). \end{aligned} \tag{6}$$

Now we combine the (1) and (4), we can rewrite the private estimate error as:

$$e_{i,t}^P = \varepsilon_t + \delta_{i,t}^I + \delta_{i,t}^R. \tag{7}$$

Where ε_t indicates the uncertainty that cannot be eliminated by the observable signals, $\delta_{i,t}^I$ stems from the incompleteness of local agent information, while $\delta_{i,t}^R$ arises from the discrepancy between the actual learned reasoning of the agent and the ideal local relationship. For individual agents, the inaccuracy of their estimations may arise jointly from the unobservability of some information and the imperfection of their own local reasoning. Since agents possess heterogeneous information sets and independently learned reasoning maps, their private cognition and estimation errors are generally heterogeneous. This motivates the central question of this work: whether agents can exploit their neighbors' partial but complementary cognition to improve their estimates and understanding of the common global quantity without exchanging raw observations or reasoning models.

III. SELF-AWARE SOCIAL CONSULTATION

A. Private Performance and Self-Awareness

In the previous section, we discussed that agents can learn reasoning maps from heterogeneous information sets and thereby conduct private estimations. However, considering the subsequent information exchange with other agents, each agent needs to have self-awareness of the reliability of its own past predictions, that is, the historical reliability of its own private cognition. Let $c_{i,t} \in (0, 1]$ denote the self-awareness state of agent i at time t, representing its

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confidence in the reliability of its private cognition based only on previously realized performance.

Once the actual global quantity y_t available for the agent is obtained, it will compare the private estimation with the true value, thereby evaluating the estimation ability of its own reasoning map. We define $q_{i,t}$ as the realized quality score of the private estimate at time t, after the actual quality y_t has been disclosed. This represents the credibility of this agent's private estimate at that moment.

$$q_{i,t} := \exp\left(-\frac{|e^P_{i,t}|}{\sigma_y}\right), \qquad q_{i,t} \in (0,1]. \tag{8}$$

Here, $\sigma_y > 0$ is a fixed normalization scale for the global quantity, estimated exclusively from pre-consultation historical data. The exponential mapping yields a dimensionless score of 1 for zero private error and decreases monotonically toward 0 as the normalized error increases.

Since the estimation of the global quantity y_t occurs at many moments, self-awareness is not solely determined by a single estimate and the realized quality score. Instead, self-awareness is recursively updated based on its previous confidence and the newly realized private quality.

$$c_{i,t+1} = (1-\eta)c_{i,t} + \eta q_{i,t}, \qquad 0 < \eta \leq 1. \tag{9}$$

The parameter $\eta \in (0,1]$ determines the adaptation rate of self-awareness. A smaller η value indicates that a greater weight is given to the long-term historical reliability, while a larger η value implies a greater emphasis on recent performance. At the first consultation time t_0, the initial confidence $c_{i,t_0} \in (0,1]$ is determined from pre-consultation validation performance. When y_t is updated, it will affect $q_{i,t}$, and then influence $c_{i,t+1}$ instead of $c_{i,t}$. Therefore, the causal chain at this time is $c_{i,t} \to \hat{y}^P_{i,t} \to y_t \to q_{i,t} \to c_{i,t+1}$. The pair $(\hat{y}^P_{i,t}, c_{i,t})$ represents the private belief of agent i regarding the global quantity, and also incorporates its historically assessed reliability. These contents will play a crucial role in the subsequent information exchange during the social consultation.

B. Belief-Level Communication

Agents are connected through a fixed communication network, thereby determining which other agents they can directly exchange information with. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a fixed, undirected, and connected communication graph, where $\mathcal{V}$ is the agent set defined in the previous section and $\mathcal{E} \subseteq \{\{i,j\} : i,j \in \mathcal{V}, i \neq j\}$ is the edge set. Let $\mathbf{A} = [a_{ij}] \in \{0,1\}^{N\times N}$ denote the adjacency matrix, where $a_{ij} = 1$ if $\{i,j\} \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. Since $\mathcal{G}$ is undirected, $a_{ij} = a_{ji}$ and $a_{ii} = 0$. The graph's undirected nature implies bilateral communication, but the two agents may later assign different levels of importance or preference to each other's beliefs.

For each agent, only the directly connected agents are available for one-hop consultation. $\mathcal{N}_i$ represent the direct communication neighbors of agent i.

$$\mathcal{N}_i := \{j \in \mathcal{V} \setminus \{i\} : \{i,j\} \in \mathcal{E}\}. \tag{10}$$

Because the graph is undirected, $j \in \mathcal{N}_i$ if and only if $i \in \mathcal{N}_j$. Connectivity means each agent can communicate with agents that have different information and beliefs. This also implies that no agent will be isolated from information transmission and exchange with other agents.

Communication occurs at the belief level instead of at the raw-information or model level. Specifically, agent i does not transmit its raw local observation $\mathbf{x}_{i,t}$, local information set $\mathcal{I}_i$, or reasoning map f_i. What agent i transmits is its current private estimate $\hat{y}^P_{i,t}$ and its current self-awareness based on historical performance $c_{i,t}$. Hence, we define the private belief message transmitted by agent i as $\mathbf{b}^P_{i,t}$:

$$\mathbf{b}^P_{i,t} := \begin{bmatrix} \hat{y}^P_{i,t} \\ c_{i,t} \end{bmatrix}, \tag{11}$$

The first component reflects the agent's current private estimate, while the second encapsulates its historically evaluated reliability. At this point, we should clearly understand that what we are exchanging is not the private cognition $\mathcal{C}^P_i$ itself, but rather the external manifestation of the agent's cognition. At each consultation step, agents transmit their own private belief messages to their direct neighbors. Agents do not recursively forward socially aggregated estimates within the same consultation step. The graph specifies the available communication channels. The edge structure determines which agents can exchange private beliefs, so the graph represents the capability to transmit information.

After receiving belief messages from its neighbors, agent i must decide how much weight to assign to its own private estimate and to each neighbor's estimate. This confidence-weighted aggregation derives the social-consultation rule introduced next.

C. Social Consultation and Error Properties

For social consultation, define $\mathcal{N}^+_i := \mathcal{N}_i \cup \{i\}$, which includes agent i and all its direct neighbors.

Since each transmitted belief is accompanied by its historically assessed confidence, agent i assigns greater weight to beliefs with higher self-awareness values. Hence, we define the confidence-based social weights as:

$$w_{ij,t} := \begin{cases} \dfrac{c_{j,t}}{\sum_{k\in\mathcal{N}^+_i} c_{k,t}}, & j \in \mathcal{N}^+_i, \\ 0, & j \notin \mathcal{N}^+_i. \end{cases} \tag{12}$$

The $w_{ij,t}$ is the influence assigned by the receiver agent i who got the information from source sender agent j at time t. Since $c_{j,t} > 0$, the weights are nonnegative and satisfy $\sum_{j=1}^N w_{ij,t} = 1$. Here, we clarify that an undirected communication graph does not imply symmetric weights. This is because different agents may assign different confidences and weights to different neighbors.

We define the social estimate as:

$$\begin{aligned} \hat{y}^S_{i,t} &= \sum_{j\in\mathcal{N}^+_i} w_{ij,t}\hat{y}^P_{j,t} \\ &= \hat{y}^P_{i,t} + \sum_{j\in\mathcal{N}_i} w_{ij,t}\left(\hat{y}^P_{j,t} - \hat{y}^P_{i,t}\right). \end{aligned} \tag{13}$$

The social estimate is a confidence-weighted convex combination of the private estimates available to agent i. The second form makes the consultation update explicit. Agent i uses its private estimate $\hat{y}^P_{i,t}$ as the reference and adjusts it according to the weighted discrepancies between its own belief and those received from its neighbors. A neighbor with a larger social weight $w_{ij,t}$ exerts a stronger influence on this correction. Importantly, this operation updates only the current belief of the agent; its local information set $\mathcal{I}_i$ and private reasoning map f_i remain unchanged.

We define the matrix notation as follows: Let $\mathbf{W}_t := [w_{ij,t}] \in \mathbb{R}^{N\times N}$, $\hat{\mathbf{y}}^P_t := [\hat{y}^P_{1,t}, \ldots, \hat{y}^P_{N,t}]^\top$, and $\hat{\mathbf{y}}^S_t := [\hat{y}^S_{1,t}, \ldots, \hat{y}^S_{N,t}]^\top$. Where $\mathbf{1}_N$ is the N-dimensional all-ones vector.

$$\hat{\mathbf{y}}^S_t = \mathbf{W}_t \hat{\mathbf{y}}^P_t, \qquad \mathbf{W}_t \mathbf{1}_N = \mathbf{1}_N,$$

We define the social error relation $e^S_{i,t}$ as:

$$\begin{aligned} e^S_{i,t} &:= y_t - \hat{y}^S_{i,t} \\ &= \sum_{j\in\mathcal{N}_i^+} w_{ij,t} e^P_{j,t}. \end{aligned} \tag{14}$$

Therefore, social consultation does not involve constructing or identifying a new global model. Instead, it reorganizes heterogeneous private estimation errors by exchanging and fusing information and beliefs through communication networks. Since $w_{ij,t} \geq 0$ and $\sum_j w_{ij,t} = 1$, we can have:

$$\min_{j\in\mathcal{N}_i^+} e^P_{j,t} \leq e^S_{i,t} \leq \max_{j\in\mathcal{N}_i^+} e^P_{j,t}.$$

Hence, we can promise the local convex stability of the social error, and social fusion cannot produce an estimation error outside the convex hull of the available private errors. This also means the belief fusion can not diverge beyond the local private-error envelope.

We define the social correction as $\Delta^S_{i,t}$, and we combine it with the social estimation error; it gives us the improvement condition in our social consultation mechanism:

$$\begin{aligned} \Delta^S_{i,t} &:= \hat{y}^S_{i,t} - \hat{y}^P_{i,t} \\ &= \sum_{j\in\mathcal{N}_i} w_{ij,t} \left(\hat{y}^P_{j,t} - \hat{y}^P_{i,t}\right), \\ e^S_{i,t} &= e^P_{i,t} - \Delta^S_{i,t}, \\ |e^S_{i,t}| < |e^P_{i,t}| &\iff 0 < \frac{\Delta^S_{i,t}}{e^P_{i,t}} < 2. \end{aligned} \tag{15}$$

The last equivalence holds for $e^P_{i,t} \neq 0$. Hence, consultation improves agent i's instantaneous estimate precisely when the social correction moves the private estimate toward the truth or truth value without excessive overshoot. And if $e^P_{i,t} = 0$, its private estimate is already correct; strict improvement is impossible, so the estimate is preserved only when $\Delta^S_{i,t} = 0$.

Equations (14) and (15) show that communication is not assumed to be universally beneficial for agents' estimation. Therefore, the value of social consultations depends on whether heterogeneous neighboring beliefs can improve and correct this agent's private estimate. The following section evaluates whether such systematic improvement emerges in the proposed partially informed agent population.

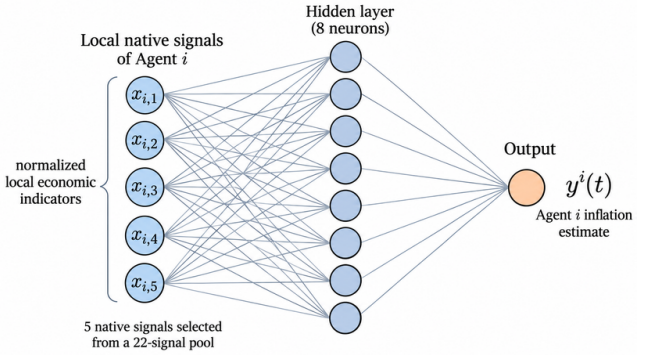


Fig. 1. Private-cognition architecture for each agent. Five local signals selected from the 22-signal pool are mapped to the private estimate $\hat{y}^P_{i,t}$ through a common 5-8-1 feedforward neural network.

IV. EXPERIMENTAL EVALUATION

A. Experimental Setup

For our proposed partial-information multi-agent framework, we use actual U.S. inflation data as the real-data testbed. The purpose is not to reconstruct the entire economic system, but to examine whether heterogeneous agents with limited local information can improve their estimation of a common global quantity through communication. In the experiment, y_t represents monthly U.S. inflation constructed from the CPI series. The global signal pool contains $M = 22$ economic indicators covering prices, labor conditions, production activity, interest rates, housing, and expectations.

We construct a population of $N = 30$ agents, each observing only $m_i = 5$ signals from the $M = 22$-signal pool, with the signal assignment balanced across the population. Thus, all agents have the same information capacity while their local information sets $\mathcal{I}_i$ differ. To isolate the effect of information heterogeneity, all agents use the same private-cognition model: a 5-8-1 feedforward neural network, illustrated in Fig. 1. The networks are trained using Bayesian-regularization backpropagation (`trainbr`) with three independent initializations. Hence, differences in private capability primarily arise from heterogeneous local information rather than unequal model complexity or training procedures.

The final private reasoning maps f_i are trained using data from January 1992 through December 2014 and are subsequently frozen throughout the social-consultation stage. Before the final fit, the 2012–2014 period serves as a held-out validation interval for estimating each agent's initial reliability; the resulting validation statistics initialize c_{i,t_0}. We then re-estimate the final private reasoning maps using all available data through 2014, before consultation begins. Initial self-awareness is set as $c_{i,t_0} = \exp(-R^{\mathrm{val}}_i)$, where R^{val}_i denotes the agent's validation NRMSE. The estimated period for the social consultation is from January 2015 to December 2017. We treat January 2018 to December 2019 as a historical out-of-sample period.

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To test whether communication and information exchange effectiveness depends on the specific communication network structure, we used four fixed connected graphs: AuthorityHub, CapabilityComplementarity, RingLattice, and Random. AuthorityHub emphasizes access to historically capable agents, CapabilityComplementarity connects agents with complementary private capabilities, RingLattice provides a decentralized regular structure, and Random serves as an unstructured baseline. We constructed these graphs under the same constraints and standards: each agent had an average of 4 neighbors, ensuring that performance differences did not arise from variations in communication density. The self-awareness adaptation rate in (9) is set to $\eta = 0.10$, and the normalization scale σ_y is computed from the pre-consultation training target. For an evaluation interval $\mathcal{T}_{\text{ev}}$, we measure individual estimation accuracy using

$$R_i^{\chi} := \frac{\sqrt{T_{\text{ev}}^{-1} \sum_{t \in \mathcal{T}_{\text{ev}}} (e_{i,t}^{\chi})^2}}{s_y(\mathcal{T}_{\text{ev}})}, \qquad \chi \in \{P, S\},$$

where R_i^{χ} denotes the individual NRMSE, with $\chi = P$ for the private estimate and $\chi = S$ for the social estimate. Moreover, $s_y(\mathcal{T}_{\text{ev}})$ denotes the sample standard deviation of y_t over the evaluation interval $\mathcal{T}_{\text{ev}}$ and serves only as the normalization scale. A smaller R_i^{χ} indicates better estimation accuracy. We additionally report the mean and median individual NRMSE, population-mean NRMSE, fraction of agents improved by consultation, and the coefficient of variation of individual NRMSE as a measure of cognitive inequality. For population-level evaluation, let $\bar{R}^{\chi} := N^{-1} \sum_{i=1}^{N} R_i^{\chi}$ denote the mean individual NRMSE, and let s_R^{χ} denote the sample standard deviation of $\{R_i^{\chi}\}_{i=1}^{N}$ across agents. We define the cognitive-inequality coefficient of variation as $\text{CV}^{\chi} := s_R^{\chi} / \bar{R}^{\chi}$, where $\chi \in \{P, S\}$. A smaller CV^{χ} indicates lower relative cross-agent dispersion in estimation performance. Here, $\bar{R}^{\chi}$ is the mean individual NRMSE and CV^{χ} measures the relative cross-agent dispersion of estimation performance, which we use as a measure of cognitive inequality. The population-mean estimate and its NRMSE are $\bar{y}_t^{\chi} := N^{-1} \sum_{i=1}^{N} \hat{y}_{i,t}^{\chi}$ and

$$R_{\text{pop}}^{\chi} := \frac{\sqrt{T_{\text{ev}}^{-1} \sum_{t \in \mathcal{T}_{\text{ev}}} (y_t - \bar{y}_t^{\chi})^2}}{s_y(\mathcal{T}_{\text{ev}})}.$$

We also report the fraction of improved agents, $p_{\text{imp}} := N^{-1} \sum_{i=1}^{N} \mathbf{1}\{R_i^S < R_i^P\}$.

B. Heterogeneous Private Cognition

Although every agent observes exactly five local signals, their information compositions differ across the population. As shown in Fig. 2, the assignment is balanced: each of the 22 global signals is observed by either six or seven agents. Thus, individual information capacity is identical and population-level signal coverage is approximately uniform, while the local information sets $\mathcal{I}_i$ remain heterogeneous.

The heterogeneous local views nevertheless produce meaningful local private cognition. As summarized in Table II, the mean and median individual private NRMSEs over

TABLE I
EXPERIMENTAL CONFIGURATION

Setting	Value
Agents N	30
Global signals M	22
Signals per agent m_i	5
Private model	5–8–1 feedforward NN
Initial Training	1992–2011
Reliability validation	2012–2014
Final private-model refit	1992–2014
Consultation evaluation	2015–2017
Historical OOS	2018–2019
Self-awareness rate η	0.10
Mean graph degree	4

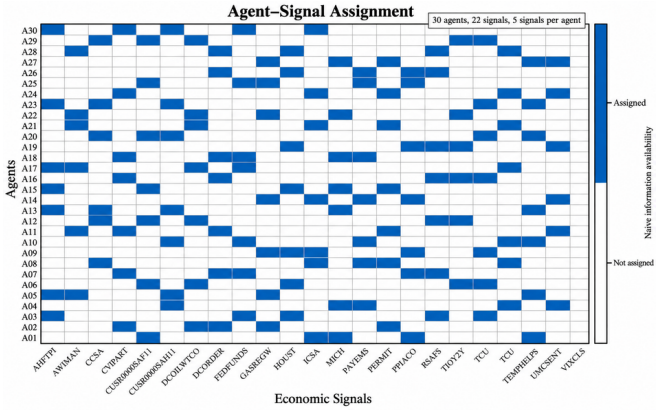


Fig. 2. Balanced heterogeneous signal assignment in Phase I. Each of the 30 agents observes exactly five signals from the 22-signal global pool, while each global signal is covered by six or seven agents.

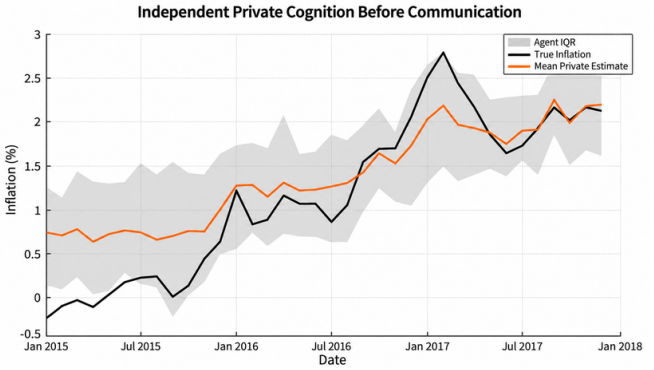


Fig. 3. Population-level private cognition before communication. The shaded region denotes the interquartile range of the 30 private estimates, the black curve is the true inflation, and the orange curve is the population-mean private estimate.

2015–2017 are 0.8212 and 0.8155, respectively, while 96.7% of the agents outperform the historical-mean baseline. The population-mean private estimate, $\bar{y}_t^P := N^{-1} \sum_{i=1}^{N} \hat{y}_{i,t}^P$, achieves a substantially lower NRMSE of 0.4053 and an R^2 of 0.7532. Fig. 3 further shows that the population mean follows the overall inflation trajectory despite the dispersion among individual private beliefs.

At the individual level, we can observe significant differences in the private capabilities of different agents. Fig. 4 provides a very clear illustration of the private estimates of all the agents at this point. The standard deviation of individual NRMSE is 0.3140, corresponding to a cognitive-

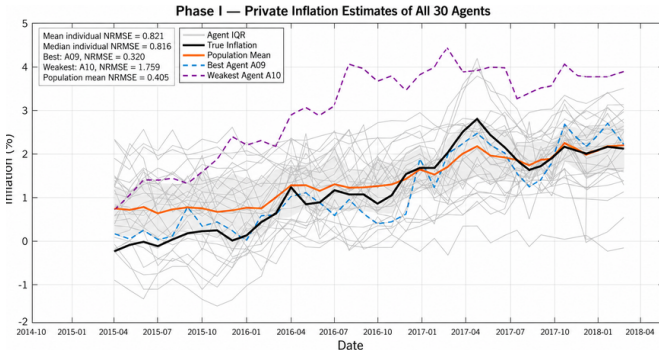


Fig. 4. Private inflation estimates of all 30 agents over 2015–2017. Despite identical information capacity and model architecture, substantial heterogeneity emerges across individual agents. A09 and A10 denote the strongest and weakest realized private agents, respectively.

TABLE II

PHASE-I PRIVATE-COGNITION PERFORMANCE OVER 2015–2017

Item	Value	Description / Native Signals
Population-level performance		
Mean individual NRMSE	0.8212	Average private error across 30 agents
Median individual NRMSE	0.8155	Median private error
Std. of individual NRMSE	0.3140	Cross-agent dispersion
Population-mean NRMSE	0.4053	NRMSE of $\bar{y}_t^P$
Population-mean R^2	0.7532	Population-level goodness of fit
Agents beating historical-mean baseline	96.7%	Fraction with useful private predictive capability
Cognitive-inequality CV	0.3824	Relative dispersion of individual NRMSE
Confidence-error correlation	-0.3406	Higher confidence tends to accompany lower error
Representative realized agents		
A09 (strong)	0.3201	GASREGW, HOUST, PPIACO, TCU, ICSA
A08 (median)	0.8368	TEMPHELPS, ICSA, PAYEMS, CCSA, PERMIT
A10 (weak)	1.7589	UMCSENT, FEDFUNDS, PAYEMS, TEMPHELPS, CUSR0000SAH1

inequality coefficient of variation of 0.3824. The strongest realized agent, A09, obtains an NRMSE of 0.3201, whereas the weakest agent, A10, reaches 1.7589. The two agents observed the same amount of information, but because the information differed in content, the quality of their private estimates differed substantially.

These illustrative cases do not establish a causal hierarchy among the individual indicators. Instead, they demonstrate that distinct combinations of individually informative signals can produce substantially divergent private estimation capabilities over the same evaluation period. Therefore, we observe that, at this point, the limitations of a single agent conducting partial reasoning with limited information are evident, and estimation capabilities differ across agents. This motivates the social-consultation experiment examined next.

C. Effect of Social Consultation

Through previous experiments, we have also discovered that partial reasoning based solely on incomplete data and private estimation have limitations. Here, we enable agents to exchange beliefs within the communication graph to validate our social consultation framework. To isolate the effect of communication, all Phase-I reasoning maps f_i and local information sets $\mathcal{I}_i$ are kept fixed throughout Phase II-A. Agents neither acquire additional raw signals nor retrain their private reasoning models. The only added mechanism is the belief-level exchange and confidence-weighted social consultation defined in Section III. The resulting cognitive change under the Authority Hub topology is illustrated in Fig. 5.

Fig. 5 provides a direct visual comparison between independent private cognition and social consultation. As can be seen from the figure, before communication, $\hat{y}_{i,t}^P$ was highly scattered and very different, which indicates the differences in private cognition and estimation ability brought about by different information combinations; After conducting social consultations, it can be observed that $\hat{y}_{i,t}^S$ has significantly contracted, and its ability to track the true value has also improved. This indicates that internal agents within the communication network share and exchange information.The social estimates become substantially more concentrated, while their tracking of the true inflation trajectory also improves. This visual contraction is accompanied by a substantial quantitative reduction in estimation error.

During the experiments from 2015 to 2017, all our statistical indicators improved significantly. Compared with the original private estimate, the Mean individual NRMSE decreased from 0.8212 to 0.4343, a relative decrease of 47.1%. The Median decreased from 0.8155 to 0.4393, the Population-mean NRMSE decreased from the original 0.4053 to 0.2908, the Individual NRMSE standard deviation decreased from 0.3140 to 0.0720, and the Cognitive-inequality CV decreased from 0.3824 to 0.1657. Under this mechanism, all agents' estimation capabilities improved compared with the previous private estimate. The corresponding quantitative comparison is summarized in Table III.

To verify the universality of our results and mechanism, we hope they hold for different communication graphs as well. We next examine whether the observed improvement depends critically on a particular communication topology. Fig. 6 compares the four fixed graph structures. We take the private estimate as the current baseline. The private baseline mean NRMSE is 0.8212. In AuthorityHub, this indicator dropped to 0.4343; under the CapabilityComplementarity structure, it dropped to 0.4566; in the RingLattice, it dropped to 0.4663; and in the randomly generated graph, it dropped to 0.5040. Among the four topological structures, agents' estimation accuracy relative to the overall system after information exchange improves significantly over the baseline. Therefore, topology affects the magnitude of the benefit, but not the existence of the communication advantage in these experiments.

We also hope to verify that the improvement in estimation effectiveness under this mechanism does not occur only during 2015-2017. The enhancement in estimation ability should also exist in the historical out-of-sample period (2018-2019). In this Historical OOS of AuthorityHub, the Mean

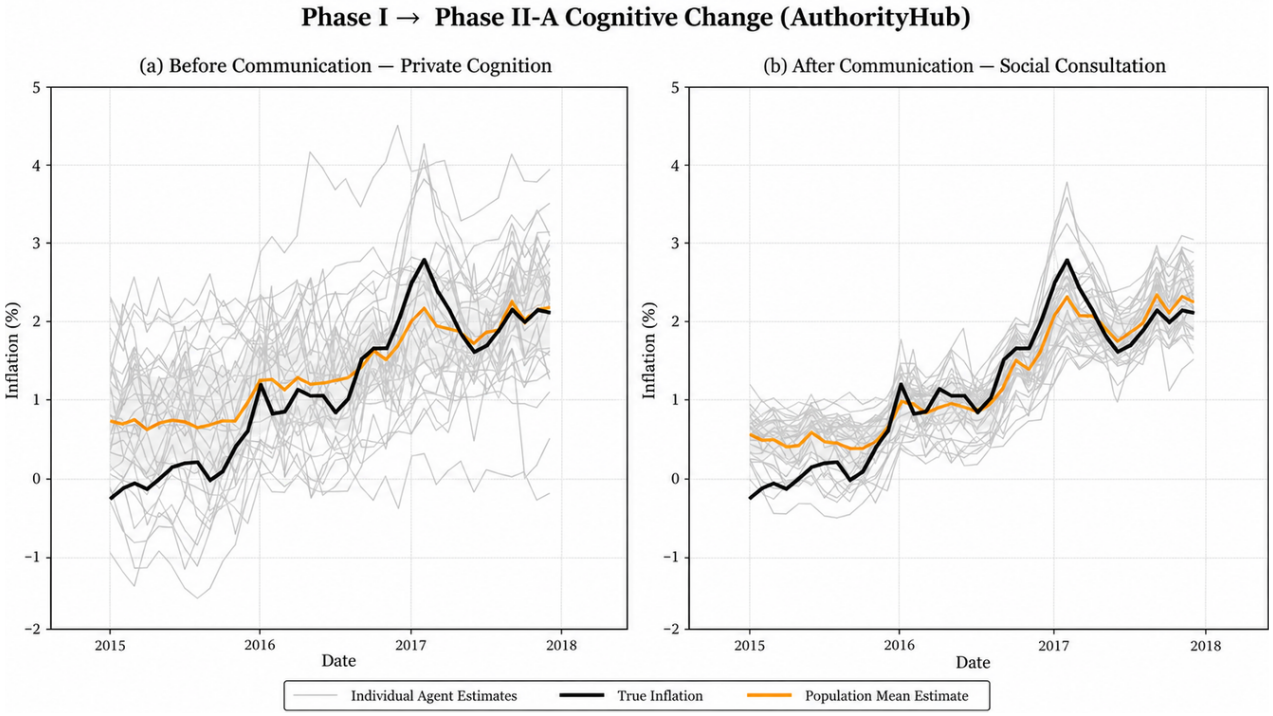


Fig. 5. Phase-I to Phase-II-A cognitive change under the AuthorityHub topology. Left: independent private cognition before communication. Right: social cognition after confidence-weighted belief exchange. Private reasoning maps and local information sets remain fixed between the two stages.

TABLE III
PHASE-II-A SOCIAL-CONSULTATION PERFORMANCE

Configuration	Mean NRMSE	Median NRMSE	Population-mean NRMSE	Agents Improved	Cognitive-inequality CV
2015–2017 development period					
Private baseline	0.8212	0.8155	0.4053	–	0.3824
AuthorityHub	**0.4343**	0.4393	**0.2908**	100%	**0.1657**
CapabilityComplementarity	0.4566	0.4641	0.3078	93.3%	0.2022
RingLattice	0.4663	0.4554	0.3200	96.7%	0.2168
Random	0.5040	0.4621	0.3412	96.7%	0.3159
2018–2019 historical out-of-sample period					
Private baseline	0.8350	0.5775	0.2034	–	0.7263
AuthorityHub	**0.4000**	**0.3952**	**0.1586**	100%	**0.3217**

individual NRMSE decreased from 0.8350 to 0.4000, the Median decreased from 0.5775 to 0.3952, the Population mean decreased from 0.2034 to 0.1586, the standard deviation decreased from 0.6065 to 0.1287, and the Cognitive-inequality decreased from 0.7263 to 0.3217. The consultation effect persists beyond the development period. The same qualitative improvement persists over the 2018–2019 historical out-of-sample period, as reported in Table III. Having established the population-level effect, we next examine how consultation benefits are distributed across individual agents and how they relate to network structure.

D. Individual Benefit and Network Robustness

We define the period-level social gain of agent i as $\Lambda_i^S := R_i^P - R_i^S$, where $\Lambda_i^S > 0$ indicates an improvement after consultation. Fig. 7 summarizes the individual-level effect. The left panel shows the substantial heterogeneity in private generalization before communication, whereas the right panel compares each agent's private and social NRMSE under AuthorityHub. All 30 agents fall below the identity line, yielding $p_{\text{imp}} = 100\%$. Moreover, $\rho_{\text{gain}} := \text{Corr}(R_i^P, \Lambda_i^S) = 0.9822$, indicating a strong association between weaker private cognition and larger consultation gains.

The benefit is also robust to communication structure. As shown in Fig. 6 and Table III, at least 93.3% of agents improve under every tested topology, while the mean social gain $\bar{\Lambda}^S := N^{-1}\sum_{i=1}^N \Lambda_i^S$ ranges from 0.3171 to 0.3869. Moreover, all four social configurations satisfy $\text{CV}^S < \text{CV}^P = 0.3824$. Hence, network structure affects the magnitude and distribution of the benefit, but broad individual-level improvement is not tied to a single topology. These results support the central premise that heterogeneous partial cognition can be converted from an individual limitation into a collective resource through belief-level communication.

Overall, after social consultation, almost all agents' estimation capabilities improve. Moreover, agents with a larger

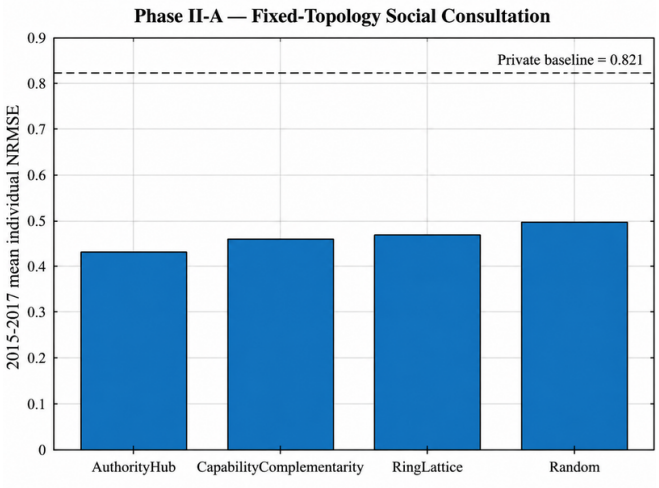


Fig. 6. Mean individual NRMSE under four fixed communication topologies over 2015–2017. The dashed line denotes the Phase-I private baseline. All four structures improve upon the private baseline, while AuthorityHub yields the lowest observed mean-agent NRMSE among the tested graphs.

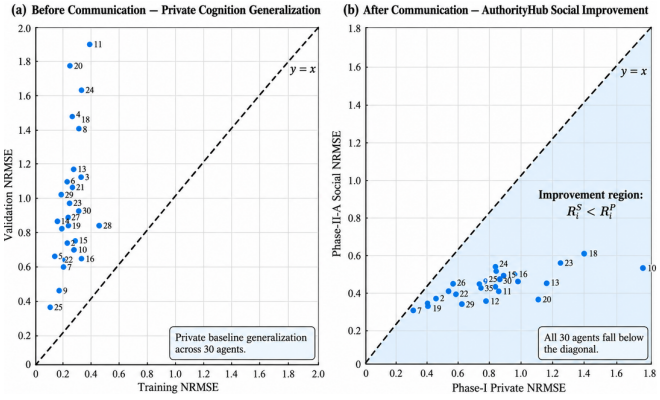


Fig. 7. Individual cognitive improvement before and after communication. (a) Private-cognition generalization across the 30 agents before communication. (b) Private versus social NRMSE under AuthorityHub; all agents lie below the identity line $R_i^S = R_i^P$ and therefore fall in the improvement region $R_i^S < R_i^P$.

original private NRMSE usually obtain a greater social gain, meaning agents with weaker initial capabilities receive a greater benefit. Second, the experimental results show that this framework and mechanism are effective across different topological structures.

V. CONCLUSION

In this paper, we considered collective estimation of a complex global quantity when each agent possesses only partial information and partial reasoning. We formulated this setting and proposed a self-aware social consultation framework in which agents first form private estimates from heterogeneous local information, evaluate their reliability from historical performance, and then exchange belief-level information over a communication network. The private reasoning maps remain fixed during consultation, allowing the effect of communication to be isolated. We also characterized the local error condition under which social correction improves an agent's instantaneous estimate. Experiments using real U.S. inflation data showed that social consultation substantially improves both individual and population-level estimation. The improvement persists across different communication topologies and a historical out-of-sample period, while agents with weaker private cognition tend to obtain larger consultation gains.

These results demonstrate that heterogeneous partial cognition can serve as a collective resource rather than only an individual limitation. The present study focuses on fixed connected communication graphs, frozen private reasoning maps, and belief-level consultation. Future work will investigate communication-topology design, richer information and reasoning exchange, and their integration into broader collective-cognition frameworks under different information distributions and global estimation tasks.

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